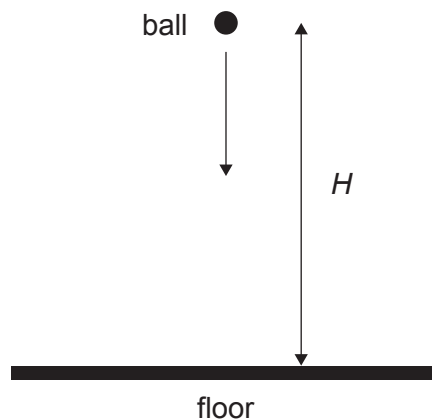
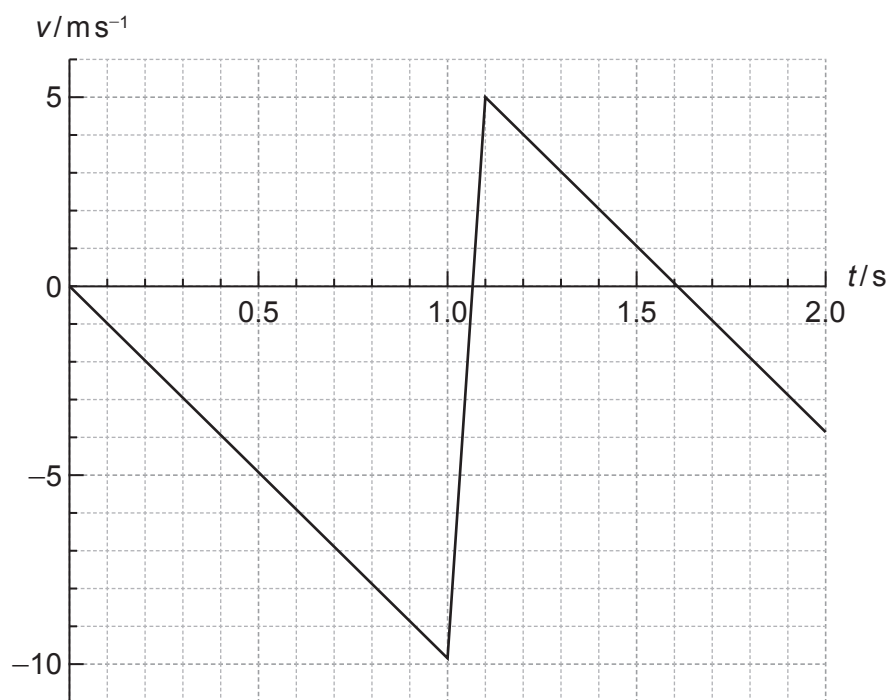


4. A ball of mass 0.250 kg is released from rest at time $t = 0$, from a height H above a horizontal floor.



The graph shows the variation with time t of the velocity v of the ball. Air resistance is negligible. Take $g = -9.80\text{ ms}^{-2}$. The ball reaches the floor after 1.0 s .



(This question continues on the following page)

(Question 4 continued)

(a) Determine H . [1]

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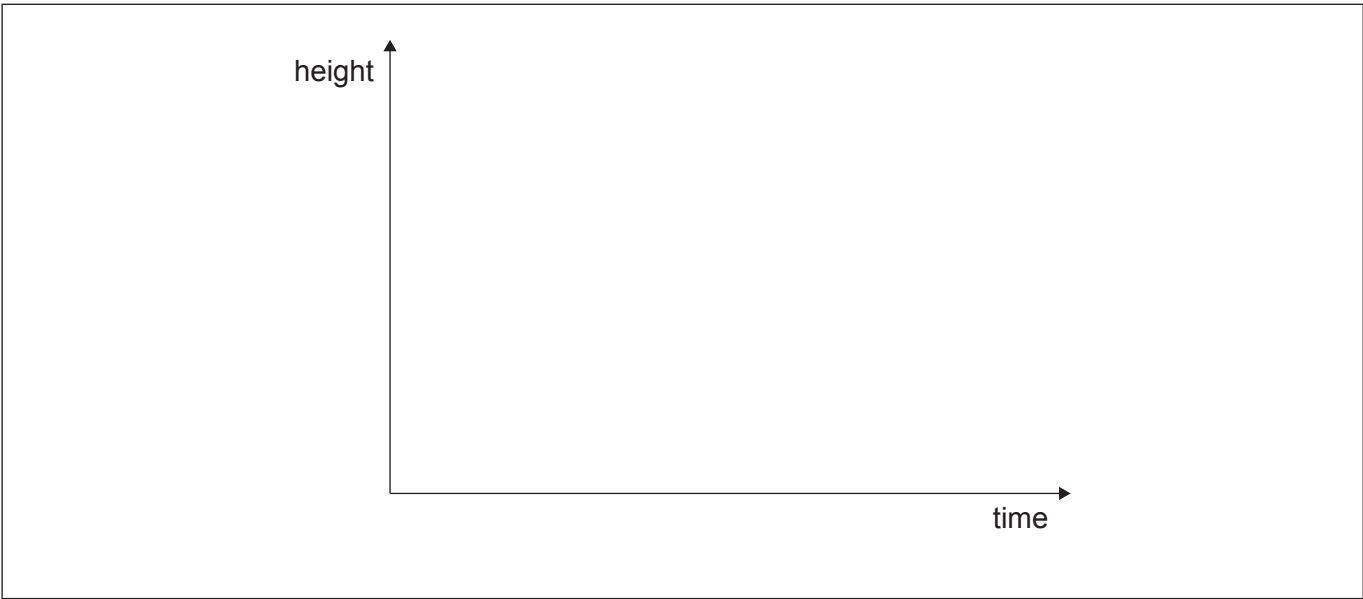
(b) (i) Label the time and velocity graph, using the letter M, the point where the ball reaches the maximum rebound height. [1]

(ii) State the acceleration of the ball at the maximum rebound height. [1]

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(iii) Draw, on the axes, a graph to show the variation with time of the height of the ball from the instant it rebounds from the floor until the instant it reaches the maximum rebound height. No numbers are required on the axes. [1]



(This question continues on the following page)

(Question 4 continued)

- (c) Estimate the loss in the mechanical energy of the ball as a result of the collision with the floor.

[1]

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- (d) (i) Determine the average force exerted on the floor by the ball.

[3]

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- (ii) Suggest why the momentum of the ball was not conserved during the collision with the floor.

[1]

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